Jinhee Mok

AI & Automation Engineer

jinheemok815@gmail.com
+82 10-9030-2888
jin-tonix.github.io
Yeoksam-dong, Gangnam-gu, Seoul

Career History

1. StyleSeller | AI & Automation PL

2025.11 – present

I automate work that staff in the business team used to repeat by hand: handling vendor inquiries, organizing records, calculating prices, and registering products. I observe the work to decide what to automate, then handle everything from gathering requirements and designing the system to implementation, deployment, and operation.

Project 1. Agent platform and skill router (Spark)

2026.05 – present

Situation

We ran the agents as several separate ones. Context was lost every time work was handed over, and tracking and managing each agent took time. Shared skills had also grown to 115, so having the LLM read the full list to pick the right skill had become slow and costly.

Action

- Designed and built the work orders and the agent delegation structure for an in-house agent platform forked from Paperclip
- Limited delegation depth to 2 levels and delegations to 5 per work order
- Enforced tool-use approval on the server and in the database, not in the prompt
- Designed a pipeline in which BM25 search first pulls out candidate skills from the full set, the judge model Jev evaluates the gate and fit, and Gemini makes the final selection
- Built a 132-question evaluation set, found errors in the answer key and fake skills, fixed them, and re-measured

Result

- Selected the correct skill in 119 of 132 evaluation cases (90.15%); the set was new when introduced and later reused during tuning
- Reduced the number of skills Gemini has to check from 115 to 5
- Reduced Gemini review from 115 skills to 5, with an estimated 90% cost reduction based on unit prices and reduced input
- Applied the improved routing structure to all sub-agents
- Made sure a tool approval is not executed twice, even with 10 concurrent requests
- Set a daily cap of KRW 30,000 on paid API call costs

Tech: Paperclip, Express, Drizzle, PostgreSQL, MCP SDK, Hermes, Python, FastAPI, LiteLLM, Redis, Langfuse, Gemini Flash

Project 2. Vendor CS agent

2026.06 – present

Situation

Vendor inquiries came in through staff KakaoTalk group chats, but there was no official API, so collecting the conversations and replying automatically was difficult. Replies often contain information that must not be wrong, such as amounts, fees, and brand names, so sending LLM-generated replies as they are was also not an option.

Action

- Built a Windows collector that gathers KakaoTalk conversations on staff PCs and passes them to the server
- Forked kakaocli and built a structure that sends replies from a staff Mac and confirms they were delivered
- Designed the agent to compare each chat room name with the list of vendors it handles, and to neither generate nor send a reply if the vendor cannot be identified
- Generated draft replies by searching vendor records and the internal knowledge hub
- Built a verification step that checks the amounts, percentages, and brand names in a draft against source material and code, and re-searches and corrects the draft if they do not match
- Improved operational stability by tracing duplicate message collection and macOS sending errors

Result

- Fixed a problem where one message was collected 51 times by adding 15-second deduplication

- Fixed a macOS accessibility freeze that made 31 of 147 sends (21%) fail, using a 3-second timeout and delivery confirmation against the KakaoTalk DB
- The agent is in operation and used for real vendor replies
- 600 commits and 107 test files for the project (as of 2026.09.23)

Tech: Swift, kakaocli, Python, LightRAG, Gemini Flash, Google Sheets, Google Drive

Project 3. Vendor status board and operations ledger

2026.10.19 – 2026.10.23 built, in operation

Situation

Records of dealings with vendors were scattered across KakaoTalk, the order mailbox, Slack, and Excel order forms attached to emails. To check the progress of each vendor, a staff member had to look through several channels by hand, and the same transaction record was often left in more than one place.

Action

- Designed the ledger data structure and the matching rules that link vendors and records
- Built the intake API and the mail collector that gather data from KakaoTalk, email, Slack, and Excel order forms
- Built 6 screen tabs for checking the progress of each vendor
- Summarized the collected records by day and displayed them on a calendar for each vendor
- Matched vendors without using an LLM, in the order of thread, business registration number, KakaoTalk room, and email
- Designed records with mismatched or uncertain matching keys to stay unclassified instead of being linked automatically, so staff check them by hand

Result

- Made the progress of about 400 vendors viewable on a single screen
- Automatically collect about 100–200 transaction records per weekday
- Built in 5 days and put into real operation
- Refresh the summaries of up to about 500 vendors every day at 18:05
- Added a cost limit that automatically stops the batch if the daily AI cost exceeds KRW 2,000

Tech: React, Tailwind CSS, Vitest, Python, Pydantic, PostgreSQL, Gemini Flash

Project 4. Internal work tools and pricing, product, and vendor matching automation

2026.03 – present

Situation

Price calculation depended on Google Sheets formulas. Staff also handled by hand the work of checking and registering products from supplier catalogs and sending customized KakaoTalk messages to sellers.

Action

- Moved the price calculation formulas from the existing Google Sheets into code, applying the same rounding units
- Separated the roles: the LLM returns only the input values that need fixing as JSON, and the code recalculates the actual amounts
- Built LLM extraction of candidate product information from product catalogs in PDF, image, Excel, and URL form
- Designed registration so that an extracted product enters the actual catalog only after a staff member confirms it
- Send customized KakaoTalk messages to the recipients received from the CRM, and switch to an approval-pending state when a send reaches 500 or more recipients
- Built a worker that performs vendor and seller matching on staff PCs

Result

- Passed 43 test cases for price calculation
- Verified the calculation results of some real products against Google Sheets, item by item
- Sent customized KakaoTalk messages to 261 people in a day (as of 2026.09.22)
- Deployed the seller-finding worker to 9 of 11 staff PCs and operate it
- Replaced the authentication key shared across PCs with an individual key for each PC

Tech: React, Vitest, PostgreSQL, PostgREST, Gemini Flash, RAG, kakaocli

Project 5. LightRAG knowledge hub and mail agent

2026.03.31 – present

Situation

Several work agents needed to refer to the same internal company information. Vendor inquiries coming into the shared mailbox also needed draft replies based on actual work materials, not just generated answers.

Action

- Designed a workspace registry and a knowledge search structure shared by several agents
- Built the sequence of search planning, judging whether search results support the answer, and verifying the draft reply
- Built a mail agent that searches internal knowledge and vendor materials for each email inquiry and creates a draft reply in the responsible staff member's Gmail drafts folder
- Built verification logic that checks up to 6 claims in a generated reply against source material
- Designed it so that a claim with no support is rewritten, and the draft is discarded if unsupported content remains after the rewrite
- Set it up so that nothing is sent automatically and the responsible staff member makes the final check

Result

- 4 agents in total share the knowledge of 13 workspaces
- Added a check so the server does not start if a workspace name is linked incorrectly
- Fixed a problem where 48 of 1,943 sent emails were missed during learning, by changing how the data cursor is saved
- Operate a mail response flow in which the responsible staff member checks the sources before sending

Tech: Python, FastAPI, Pydantic, LightRAG, PostgreSQL, pgvector, Neo4j, FlashRank, Gemini Flash, Gmail, pytest

Project 6. Content autopilot

2026.10.03 – present

Situation

About a year ago, we tried to automate content production, but because of the quality of the video generation models at that time, we could not reach full automation and ended with partial automation. Since then, the models' generation quality has improved, so we took on full automation again.

Action

- Built a process in which a staff member requests a Reel in the internal work tool, followed by script generation, video rendering, and preview approval
- Built the generation of 9:16 MP4 videos and their storage in Google Drive
- Designed a state machine, not an LLM, to manage the progress of each production stage
- Set the defaults to manual approval for posting, dry-run for recurring automation, and paid generation disabled
- Made the AI-generated content label attach to generated content automatically
- Separated the review before posting from the approval step

Result

- Completed an operational canary that generates 1080×1920 MP4 videos and stores them in Drive
- The new video production path is still improving prompt quality and is at the stage before real SNS posting
- A separate path that applies per-account captions and hashtags to existing videos and posts them to several accounts at the same time is in real operation
- Enforced a limit of one post per account per day with a DB index
- 617 test files in total across the 3 related services

Tech: Python, Gemini Flash, Hermes, HyperFrames, ffprobe, Google Drive, Supabase, PostgreSQL

Project 7. Staff PC deployment and operations environment

2026.02 – present

Situation

Several automation tasks, such as KakaoTalk collection, Instagram posting, and seller matching, had to run on staff Macs and Windows PCs. Installing and managing each PC by hand made updates and authentication difficult to manage, and the operating burden was growing with every new employee.

Action

- Built Mac and Windows installers that need only a one-line command
- Designed device-level authentication with single-use tokens, issuing an individual key to each device
- Built staggered auto-updates for Mac tools and the Windows CS tool; the separate Windows collector still requires reinstalling
- Added a safeguard so the server deploys only container images that pass pytest, and stops the deployment if there is no such image
- Checked for service problems with a Slack monitoring cron
- Built a replacement server and restored service when the internal tool API stopped after the cloud DB usage limit was exceeded

Result

- Installed and deployed on 11 staff Mac and Windows PCs in total, using the one-line command
- On 2026.08.20, when the internal tool API stopped after the cloud DB usage limit was exceeded, built the same PostgREST API on a VM and restored service the same day
- Restored the service without changing the more than 100 DB call sites inside the app
- Moved 4 DBs to a self-hosted environment between 2026.08.06 and 08.25

Tech: macOS Bash, Windows PowerShell, launchd, Windows Task Scheduler, Node.js, Python, PostgreSQL, PostgREST, NGINX, Docker, GCP, Slack

2. We:Review | AI Development Intern

2025.08 – 2025.10

Project 8. Receipt inspection automation pipeline

Situation

Staff were extracting and checking product names and prices from receipt images by hand. When a receipt was long or its text was scattered, the OCR result was hard to use as it was, and rechecking the products and prices took a lot of time.

Action

- Split receipt images into tiles with PaddleOCR and extracted the text
- Analyzed the scattered OCR results with DBSCAN and restored them into lines
- Applied few-shot prompting to a Qwen-family open LLM to extract product names and prices into a JSON structure
- Turned the inspection pipeline into a service with Python and FastAPI

Result

- Raised inspection accuracy from 70% to 95%
- Cut inspection time by 85% (measured directly)
- Took part in building the inspection automation pipeline that connects OCR to LLM structured data extraction, and in turning it into a service

Tech: Python, PaddleOCR, DBSCAN, Qwen-family LLM, FastAPI

3. Previous experience

Jeonwon Daycare Center and After-School Program | Operations Manager 2015.03 – 2018.02

Hiring teachers, planning the annual curriculum and events, communicating with parents and the operations committee, and managing operations on behalf of the director.

Jeongsang Language School, Pots&Pans, Montekids English Academy | English Instructor 2018.08 – 2024.02

Designing lessons, preparing lesson plans, setting study plans for each student, and counseling parents and managing students' learning.